

Kiste Phase 8.5 Spec

Stabilize Kiste for Release

One-sentence spec: Kiste Phase 8.5 stabilizes the modular self-hosting Kiste platform for release by freezing APIs, hardening CLI/library behavior, improving tests and documentation, validating bootstrap flows, checking security, and preparing versioned release artifacts.

Phase	Main focus
1	Local Docker repo scan + token inspection
2	GitHub/Hugging Face remote scan + CI testing
3	Multi-repo workspace graph + token access matrix
4	GitOps repo generation + SOPS-style encrypted secrets
4.5	Review, refactor, and architecture checkpoint
5	Python library + CLI wrapper
6	Many repo providers + Docker Hub + serverless planning
7	Data repository + dataset intelligence
8	Self-hosting + modular bootstrap + cloud/Argo CD preparation
8.5	Release stabilization, hardening, packaging, and readiness checks

1. Purpose

Phase 8.5 is the release-stabilization checkpoint after Kiste has become broad: library, CLI, repo providers, image inspection, serverless planning, data repositories, GitOps, module registry, self-hosting, cloud readiness, and Argo CD preparation.

The goal is not to add another large capability. The goal is to stop expanding temporarily and make the product installable, testable, documented, safe by default, and reliable enough for early users.

2. Release Principle

Before Kiste becomes bigger, Kiste must become reliable.

- Freeze the public API surface that users can import.
- Freeze the CLI command tree and help text for the first release candidate.
- Validate staged bootstrap from an empty folder.
- Make security guardrails enforceable, not just documented.
- Generate release artifacts, release notes, and readiness reports.

3. Phase 8.5 Scope

Included	Excluded
API and command freeze; bootstrap validation; module doctor; security hardening; test matrix; packaging; release notes; documentation freeze; release candidate checklist.	No new provider families; no full cloud provisioning; no production Argo CD sync; no real Vault integration; no automatic dataset download; no autonomous PR merge; no runtime controller; no paid SaaS platform yet.

4. Public API Freeze

Phase 8.5 should freeze import paths that early users and future integrations can rely on. Internal implementation can change, but public imports should not break without a version bump.

```
from kiste.scanner import scan_repo
from kiste.workspace import load_workspace, scan_workspace
from kiste.gitops import generate_gitops_repo, validate_gitops_repo
from kiste.secrets import SopsSecretBackend
from kiste.tokens import check_token
```

- Public functions should return typed results where possible instead of loose dictionaries.
- Library modules must not import the CLI. The CLI imports the library.
- Deprecation warnings should be used before removing public symbols.
- Public API examples should be tested in CI.

5. CLI Command Freeze

The command tree should be stable enough for documentation, demos, CI jobs, and early users. Each command should have help text, predictable exit codes, JSON output where useful, and safe defaults.

```
kiste scan
kiste check
kiste repo
kiste token
kiste workspace
kiste gitops
kiste secrets
kiste image
kiste target
kiste data
kiste module
kiste bootstrap
kiste cloud
kiste argocd
kiste self
kiste release
```

6. Release Command Group

Phase 8.5 adds a release command group focused on readiness, packaging, and release artifacts.

```
kiste release check
kiste release freeze
kiste release test
kiste release package
kiste release notes
kiste release candidate --version 0.8.5-rc1
kiste release doctor
```

Command	Purpose
release check	Run full release readiness checks and produce a pass/warn/fail report.
release freeze	Snapshot public APIs, command tree, module manifest, and expected outputs.
release test	Run or orchestrate the local test matrix: unit, CLI, API, bootstrap, security, packaging.
release package	Build wheel/sdist artifacts and validate package metadata.
release notes	Generate release notes from specs, commits, and known changes.
release candidate	Create a release candidate checklist for a specific version.

7. Bootstrap Validation

Phase 8 introduced staged bootstrap. Phase 8.5 should test every bootstrap level from an empty folder and ensure no dangerous action happens by default.

```
mkdir demo-kiste
cd demo-kiste

kiste bootstrap core
kiste bootstrap workspace
kiste bootstrap gitops
kiste bootstrap argocd
kiste bootstrap cloud
kiste bootstrap self
```

Expected bootstrap result

- [OK] Expected .kiste files created.
- [OK] Module manifest created.
- [OK] GitOps skeleton created without plaintext secrets.
- [OK] Argo CD YAML generated with safe defaults.
- [OK] Cloud plan generated but no provisioning performed.
- [OK] Self-check and bootstrap report created.

8. Security Hardening

Release stabilization must make safety rules executable. Security should be checked by commands and CI tests, not only described in docs.

Risk	Required Phase 8.5 behavior
Raw token storage	Do not store raw tokens. Prefer env:// references or approved secret backends.
Fake encryption wording	Do not call base64 or encoding encryption. Rename files or implement real encryption.
Plaintext GitOps secrets	Fail release check if plaintext Kubernetes Secret manifests or .env files appear in GitOps output.
Cloud provisioning	Cloud commands generate plans by default; apply/provision must be explicit and gated.
Argo CD production sync	No production auto-sync by default. No prune by default.
Self-improvement PRs	Kiste can generate reports and patch plans, but not auto-merge.

9. Module Stability

Phase 8 introduced a modular Kiste architecture. Phase 8.5 should make module behavior predictable and debuggable before release.

```
kiste module list
kiste module graph
kiste module doctor
kiste module enable data
kiste module disable data
```

- If a module is missing, Kiste should explain how to install or enable it.
- If a module is disabled, commands should fail gracefully with a clear message.
- If a dependency is missing, Kiste should report the dependency chain.
- Module loading should avoid importing heavy provider SDKs until needed.

10. Test Matrix

Phase 8.5 requires tests that reflect how real users will install, import, run, bootstrap, and package Kiste.

Test group	Examples
Unit tests	Models, parsers, planners, provider adapters, graph logic.
Library API tests	Public imports and function behavior.
CLI tests	--help, exit codes, JSON output, smoke commands.
Bootstrap tests	core, workspace, GitOps, Argo CD, cloud, self from empty folder.
Security tests	token storage, plaintext secret detection, destructive-command guardrails.
GitOps tests	Kustomize structure, secret paths, validation output.
Argo CD tests	Application and AppProject YAML schema/safe defaults.
Cloud plan tests	Plan output only; no provisioning by default.
Data/image tests	Mock providers, image reference parsing, no large download by default.
Packaging tests	Build wheel/sdist and validate metadata.

```
python -m pytest -q
python -m ruff check .
python -m mypy src/kiste
python -m build
twine check dist/*
```

11. Documentation Freeze

Phase 8.5 should prepare release documentation with copy-paste runnable examples. Each major workflow should be represented by one minimal example.

```
docs/  
  quickstart.md  
  install.md  
  commands.md  
  library-api.md  
  bootstrap.md  
  security.md  
  gitops.md  
  argocd.md  
  cloud-plan.md  
  data-providers.md  
  release-notes.md
```

Required workflow examples

- Local repo scan.
- GitHub repo scan.
- Workspace graph.
- GitOps generation.
- SOPS-style secret check.
- Docker image inspect.
- Target plan.
- Data repo inspect.
- Bootstrap Argo CD.
- Self check.

12. Release Readiness Reports

Kiste should write structured reports under `.kiste/release/` so release readiness can be reviewed by humans and consumed by CI.

```
.kiste/release/  
  release-check.json  
  release-readiness.md  
  api-freeze.md  
  command-freeze.md  
  test-report.md  
  security-report.md  
  packaging-report.md  
  docs-report.md  
  release-notes.md  
  release-candidate-checklist.md
```

13. Example Release Readiness Output

Kiste Release Readiness

API:

[OK] Public imports available

[WARN] Some functions still return raw dicts instead of typed models

CLI:

[OK] Main commands available

[WARN] data commands need more examples

Tests:

[OK] unit tests pass

[OK] CLI smoke tests pass

[WARN] provider tests use mocks only

Security:

[OK] no plaintext GitOps secrets

[FAIL] token store still allows base64-encoded raw token storage

Bootstrap:

[OK] bootstrap core works

[OK] bootstrap workspace works

[OK] bootstrap gitops works

[OK] bootstrap argocd works

[WARN] cloud credential checks are partial

Release:

[OK] pyproject version found

[WARN] README needs installation troubleshooting section

14. Release Levels

Version label	Meaning
v0.8.5-dev	Internal development build.
v0.8.5-alpha	Feature-complete but unstable. For testing by the Kiste team.
v0.8.5-beta	Usable by early users. APIs mostly stable.
v0.8.5-rc1	Release candidate. Only bug fixes and docs fixes should be allowed.
v0.9.0	First serious public/private release.

Recommended Phase 8.5 target: v0.8.5-rc1 or v0.9.0-alpha, depending on how public the release should be.

15. Release Checklist

- [] pyproject version updated.
- [] license file included.
- [] README updated.
- [] quickstart tested from clean environment.
- [] CLI --help works.
- [] public Python imports work.
- [] tests pass.
- [] lint passes.
- [] type check passes.
- [] package builds.
- [] package metadata valid.
- [] release notes generated.
- [] no raw tokens in fixtures.
- [] no plaintext secrets in GitOps examples.
- [] destructive commands guarded.
- [] bootstrap flows tested.
- [] cloud and Argo CD commands generate only plans/manifests.
- [] self-improvement does not auto-merge.

16. Acceptance Criteria

Phase 8.5 is complete when these commands work and produce structured release artifacts.

```
kiste release check
kiste release freeze
kiste release test
kiste release package
kiste release notes
kiste release candidate --version 0.8.5-rc1
```

Required outputs

- [OK] release readiness report.
- [OK] API freeze report.
- [OK] command freeze report.
- [OK] test report.
- [OK] security report.
- [OK] docs report.
- [OK] packaging report.
- [OK] release notes.
- [OK] release candidate checklist.

17. Final Direction

Phase 8.5 is the stop building, make it releasable phase.

The final principle is simple: Kiste should be installable, understandable, testable, safe, and ready for early users before Phase 9 adds more powerful production capabilities.